Hanoi Urban Noise Mapping

Data Collection Report

363 smartphone measurements · 3 sites · 10 Jun to 22 Jul 2026

363	3	47-88	0.25	39%
Measurements	Sites	L_A,25s range	Model R^2	> QCVN day

1. Objective

Map and characterise urban noise in three contrasting districts of Hanoi, and predict noise from urban morphology. We follow the Sunbird AI methodology (Urban Noise Uganda 61K, Nsumba et al., 2026): smartphone field collection plus a morphology-to-noise model, reproduced then applied here.

Measurement status: our target is a 20-30 s A-weighted level from consumer smartphones (denoted L_A,25s), not a certified L_Aeq. The three phones are cross-calibrated against each other, never against a reference instrument. CONTRASTS between places and hours are supported, ABSOLUTE levels are indicative (plausible bias +3.0 to +7.3 dB, section 3.4). No compliance claim is made anywhere in this report.

2. Data collection summary

Site	n	Median dB	Min-Max	% < 60 dB	% roadside
Ocean Park	184	68	47-88	28%	73%
Hoan Kiem lake	99	70	55-80	7%	89%
Vinh Tuy area	80	66	53-76	20%	74%
ALL	**363**	**68**	**47-88**	**21%**	**78%**

Zones: Ocean Park = new high-rise development (shielding, construction); Vinh Tuy = transport infrastructure (heavy traffic); Hoan Kiem = historic old quarter (narrow streets, pedestrian zones).
QC: site labels cross-checked against GPS (6 mislabelled submissions reassigned).

Headline findings: median 68 dB; 39% of short samples sit above the QCVN daytime threshold value; levels peak at 18:00. A model trained directly on our data reaches R^2 0.25 [0.10, 0.34] under buffered loo 300 m (section 5), against baselines reported in the same table.

3. Methodology

3.1 Field data-collection protocol

Study zones were mapped in Google Earth and chosen for contrasting typologies (section 2). At each point we recorded a 20-30 s reading (with a 10 s audio clip) in ODK Collect, submitted live to a KoboToolbox server with automatic timestamp and GPS. Phones were held at 1.2 m (SPB positioning) at varied distances from the road, by three cross-calibrated collectors, from 05:00 to 23:00 with emphasis on rush hours. Weather (Open-Meteo) and spatial context (OSM) are added.

3.2 Reproduction of the Sunbird pipeline

We reproduced the full Sunbird chain on their Uganda dataset (notebooks 01-06): cleaning, audio QC, morphology features, figures, surrogate model. Our statistics match the paper (median 49 dB vs their 45-50 dB).

3.3 Model features

Morphology within 300 m of each point (OpenStreetMap): built-area ratio, road density, intersection count, distance to nearest road; plus hour of day and weekend flag. Model: LightGBM.

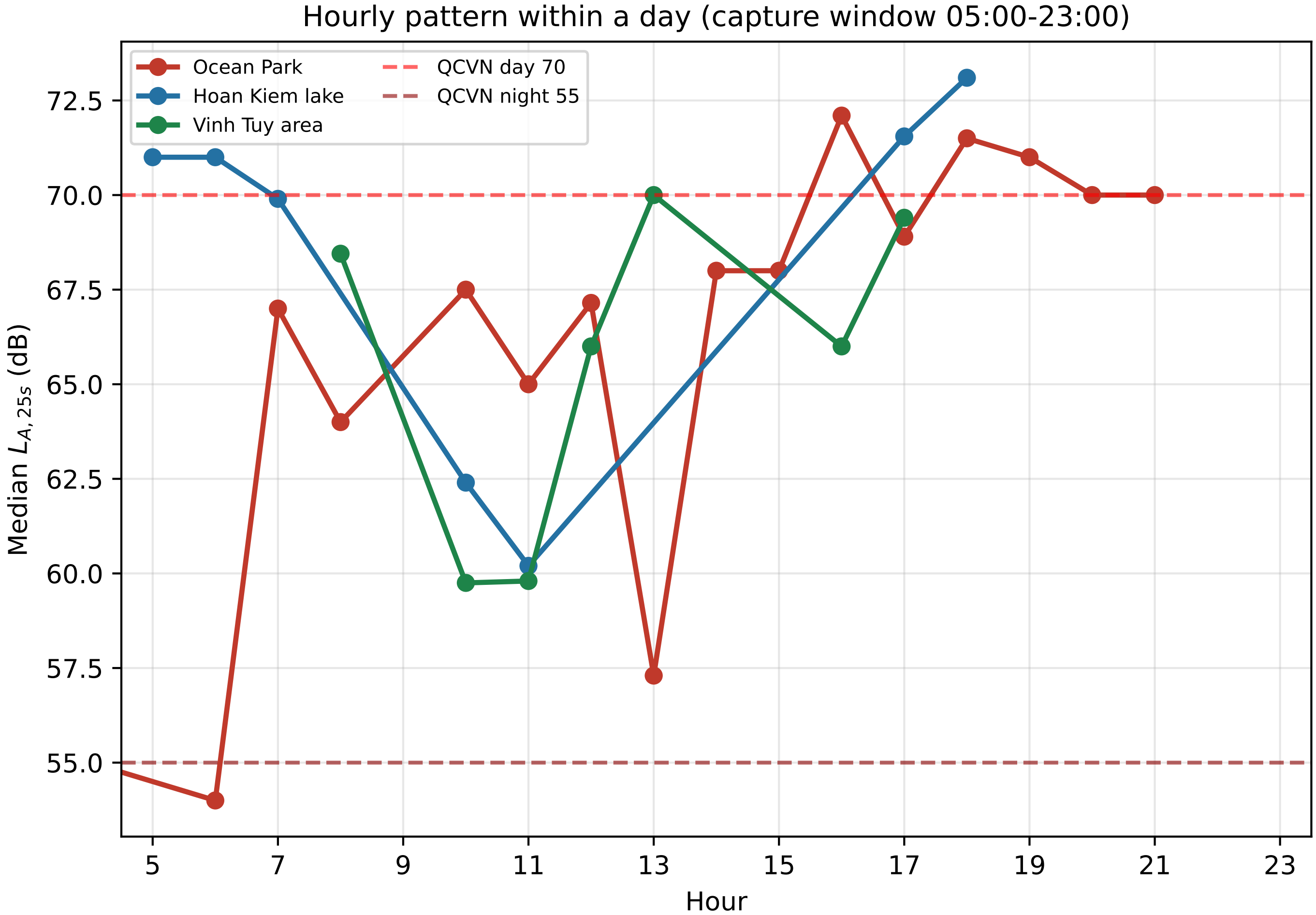
3.4 Reference values and measurement status

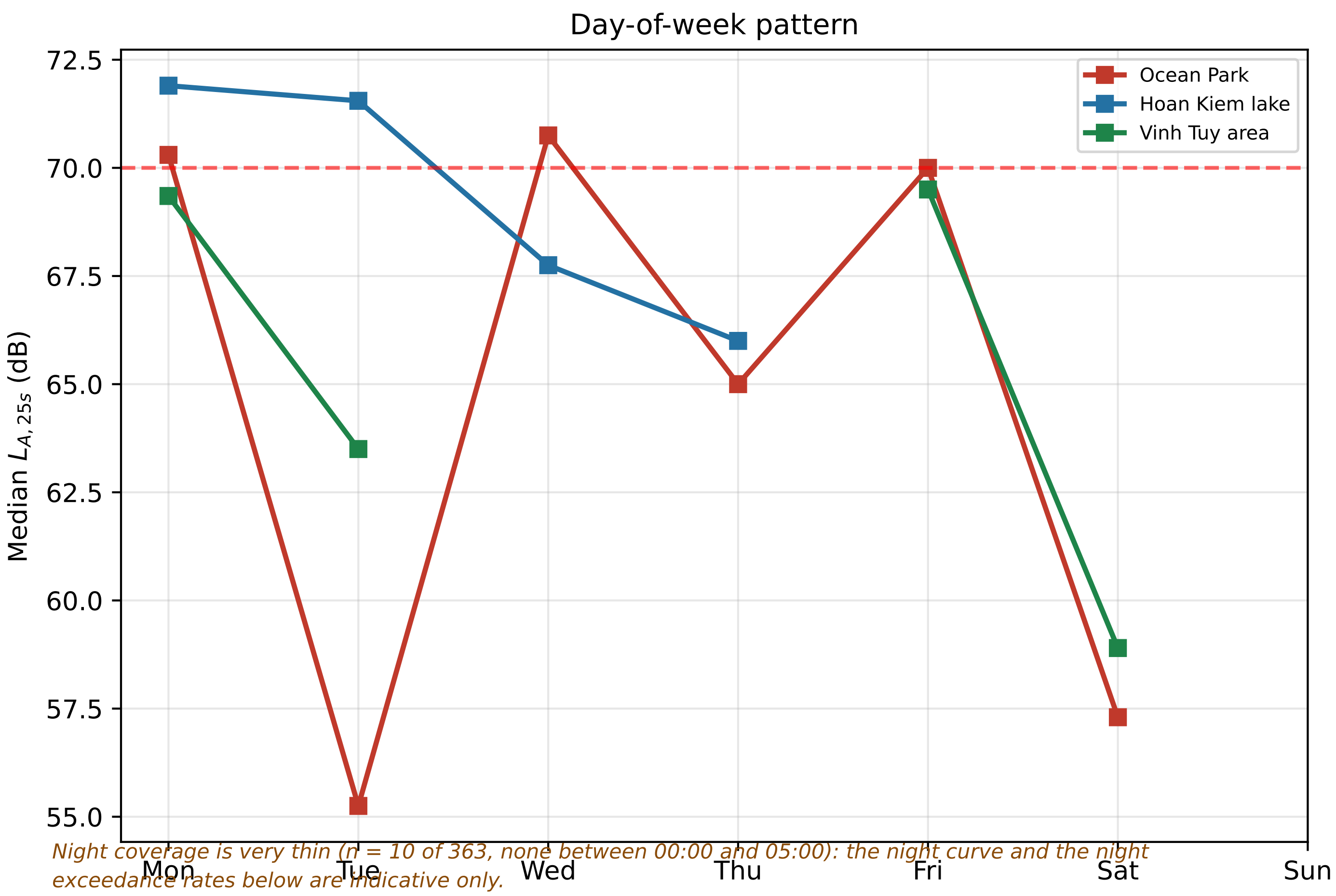
Reference value	Day (6-21h)	Night (21-6h)
QCVN 26:2010/BTNMT (Vietnam, ordinary area)	70 dB	55 dB

The WHO road-traffic guideline values (53 / 45 dB) have been WITHDRAWN from this report. They are L_den and L_night: ANNUAL averages with +5 dB evening and +10 dB night penalties. Comparing them to a 25 s daytime sample compares different statistics of different processes. QCVN is retained because it regulates a level, but it regulates an L_Aeq measured with a class 1-2 meter under TCVN 7878-2:2010: our instrument does not meet that specification either.

Consequence for every exceedance figure in this report: it is a DESCRIPTIVE STATISTIC of our sample - the share of short samples above a threshold value - and never a finding of regulatory non-compliance. Sensitivity to the calibration bias (+3.0 to +7.3 dB) is given on page 4.

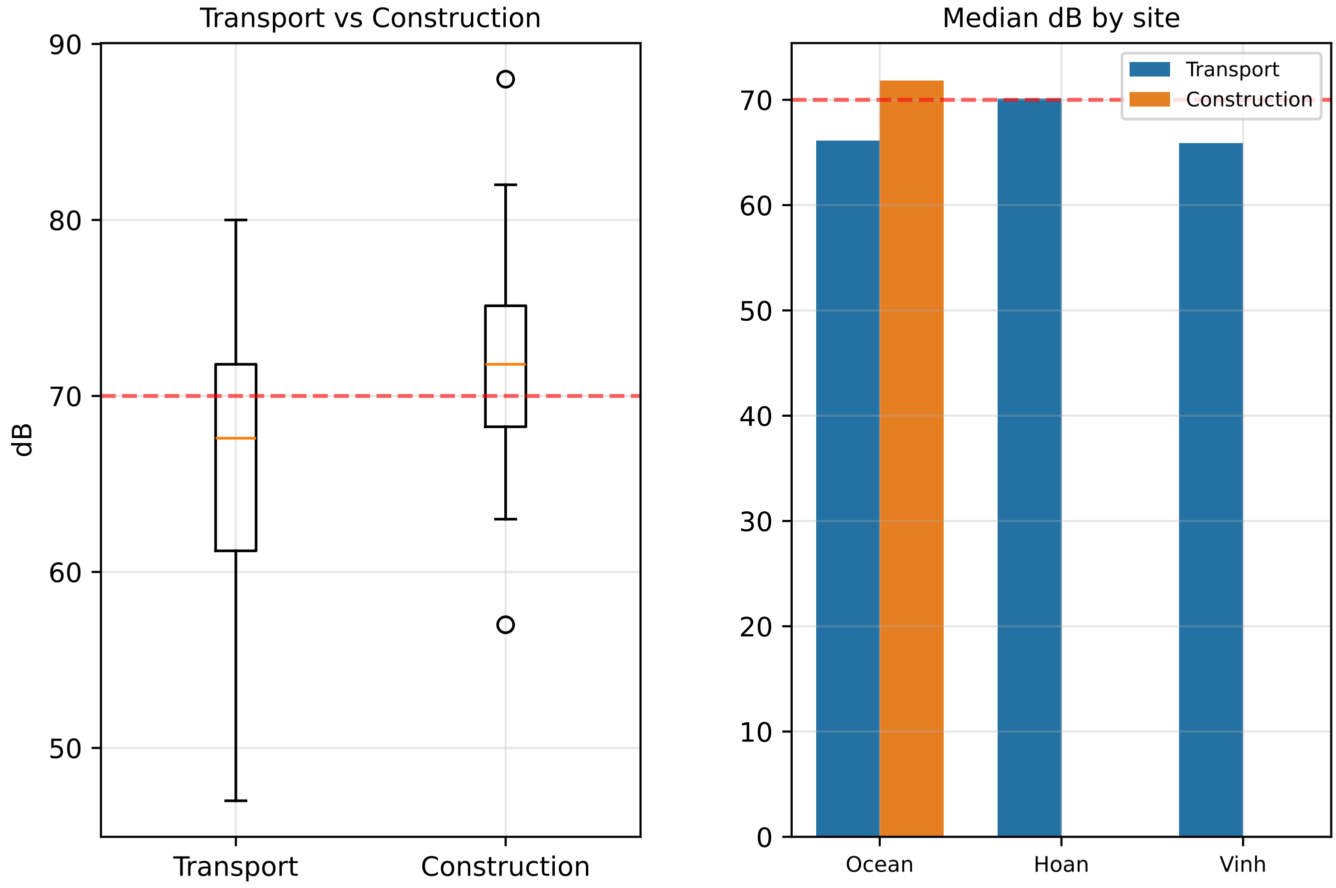
4. Data analysis: temporal patterns

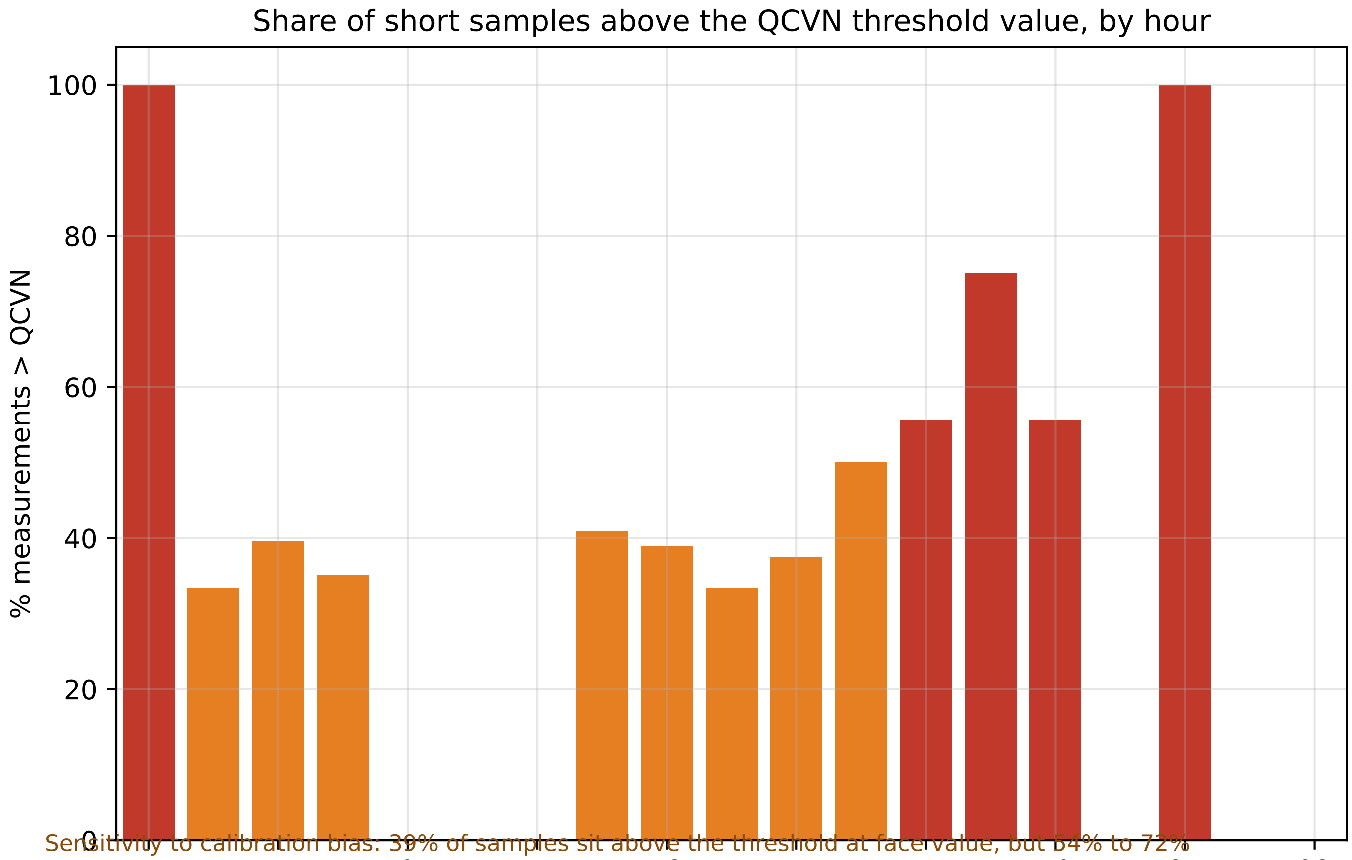




Night coverage is very thin (n = 10 of 363, none between 00:00 and 05:00): the night curve and the night exceedance rates below are indicative only.

4. Data analysis: sources & exceedances

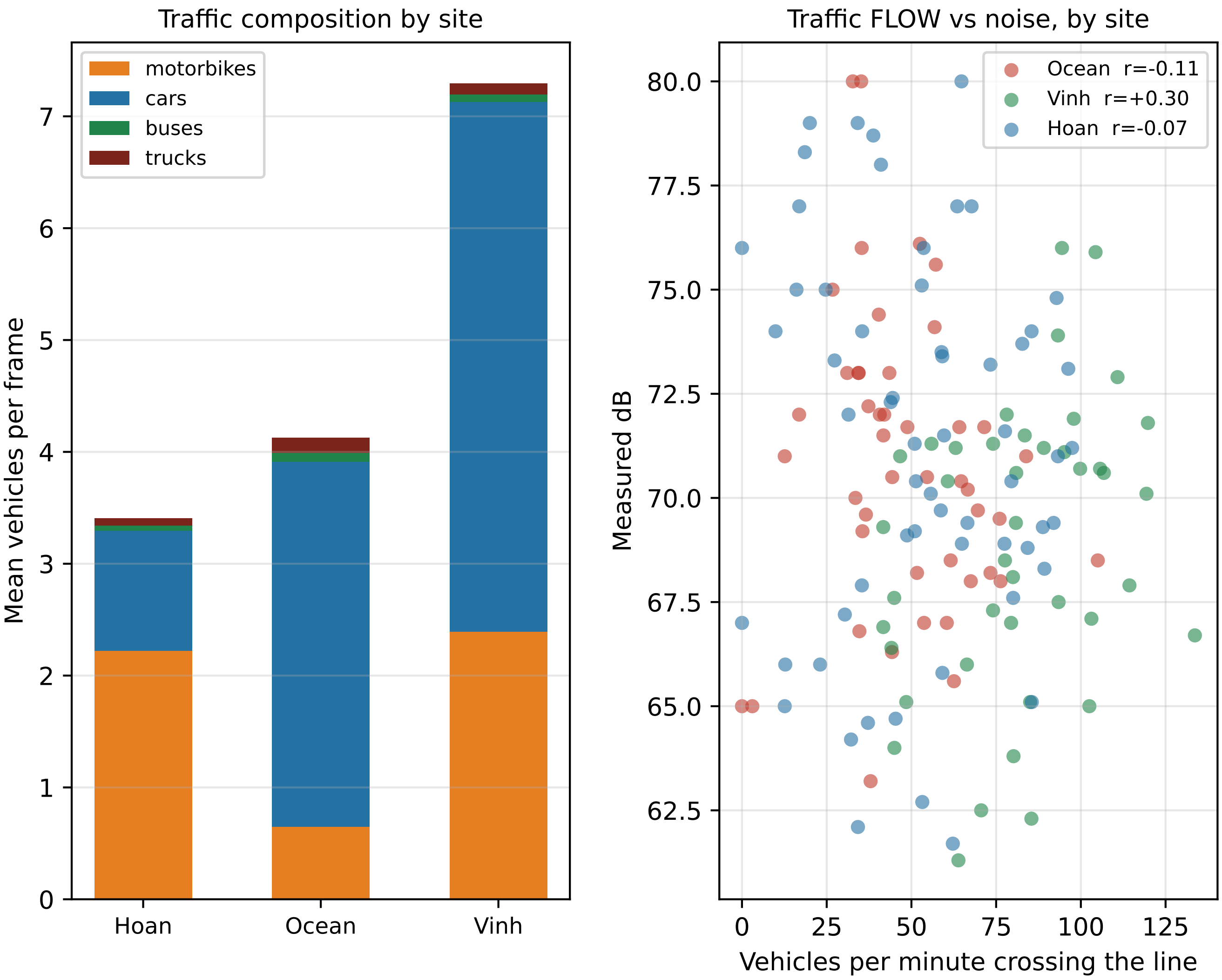




Sensitivity to calibration bias: 39% of samples sit above the threshold at face value, but 54% to 72% once the plausible instrumental bias (+3.0 to +7.3 dB, section 3.4) is applied. The exposure PATTERN - which hours and which sites are loudest - is unaffected by a common bias; the exceedance LEVEL is. We report the pattern as a finding and the level as an indication.

4. Data analysis: traffic videos & weather

147 traffic videos (~25 s each) matched to their measurement by timestamp (median gap 15 s). YOLOv8 + ByteTrack at ~10 frames/s (36572 frames): vehicles are TRACKED and counted when they cross a virtual line, giving a real FLOW in vehicles/min - not just a density per frame.



Findings: traffic composition matches the zone typologies: Hoan is motorbike-dominated (65%), Ocean car-dominated (79%), Vinh the densest corridor (up to 15 vehicles/frame). Mean flow is 59 veh/min (median 59, max 134).
MOVING TO REAL FLOW DOES NOT RECOVER THE SIGNAL. Correlation with level, by site:
density Hoan -0.11, Ocean -0.19, Vinh +0.22;
flow Hoan -0.07, Ocean -0.11, Vinh +0.30.
Pooled: density -0.15, flow -0.11 - still the wrong sign; motorcycle flow is uncorrelated (-0.004). Only Vinh, the through-traffic corridor, is positive, and it IMPROVES with flow (+0.22 -> +0.30) - the direction physics predicts. What is still missing is SPEED and the source-receiver DISTANCE: the field of view is not georeferenced. See section 7.
Note: small motorbikes are harder to detect than cars and parked vehicles are counted too, so motorbike shares are a lower bound; the detector was never validated against manual counts.

Weather: no robust effect. Raw correlations (temperature +0.26, rain -0.29) vanish once hour of day and session are controlled: quiet-point campaigns happened to fall on rainy days. Wind shows no effect on readings (no microphone artefact).

5. Predictive model

Seven models compared on identical splits. Delivered: Noyau physique seul (3 paramètres).

5.1 How performance is measured (corrected August 2026)

Earlier versions of this report grouped cross-validation on ~110 m cells. The model features are aggregates over a 300 m RADIUS, so two points 110 m apart share more than 85% of their disc: the model saw near-twins of its test points and the reported score was not out-of-sample. That protocol is replaced by buffered loo 300 m, and every model below - including the baselines - is scored on exactly the same splits, with bootstrap confidence intervals resampled by spatial block.

5.2 Model comparison - Buffered LOO 300 m

Model	R^2	95% CI	MAE (dB)	r
Moyenne globale	-0.04	[-0.12, -0.03]	6.06	-0.44
Moyenne par (site, heure)	-0.42	[-0.77, 0.03]	6.58	0.06
Régression sur log(distance route)	0.20	[0.08, 0.27]	5.28	0.45
Distance inverse (k=8, p=2)	-0.20	[-0.37, 0.02]	6.44	0.02
LightGBM — morphologie seule (ablation)	0.04	[-0.24, 0.23]	5.63	0.41
LightGBM — morphologie + temps (v1)	0.14	[-0.10, 0.33]	5.26	0.47
LightGBM v2 — voiries séparées + heure cyclique	0.10	[-0.13, 0.25]	5.54	0.43
Noyau physique seul (3 paramètres)	0.25	[0.10, 0.34]	5.01	0.50
HYBRIDE — physique + LightGBM sur le résidu	0.12	[-0.22, 0.29]	5.39	0.45
HYBRIDE conservateur — résidu bridé (morpho+temps)	0.14	[-0.15, 0.35]	5.30	0.45

Key figure - the ranking inverts between protocols. Under the permissive 600 m block split the elaborate models lead (hybrid physics+ML: 0.395, LightGBM v2: 0.332, physical core: 0.255). Under buffered loo 300 m - whose exclusion radius equals the feature aggregation radius - the order reverses: the 3-parameter physical core reaches 0.246, ahead of the distance regression (0.200) and of the hybrid (0.123). The learned residual is a NET LOSS here: dR2 -0.123.

5.3 Generalisation to an unseen typology (leave-one-site-out)

Test site	n	R^2	MAE (dB)
Ocean Park	184	0.20	5.90
Hoan Kiem lake	99	-0.08	4.60
Vinh Tuy area	80	0.42	3.51

Each site is predicted by a model trained on the other two only. A negative R2 means: worse than predicting the global mean everywhere. Maps are clipped to the sampled envelope regardless. Pooled over all sites the delivered model holds R2 +0.222, the hybrid falls to +0.035: the learned residual helps INSIDE sampled typologies and hurts outside them, so it is not applied to the published map.

Cross-city transfer (Uganda -> Hanoi) is reported in section 7 as a methodological result, not as a method.

6. Agent-based simulation (GAMA)

The predicted noise map is loaded into an agent-based model with moving vehicles, construction sites and interactive scenario controls.

6.1 What the simulation contains

Layer	Source	Status
Background noise level, per hour	LightGBM on our 363 measurements	predicted
Traffic density and vehicle mix	147 videos, YOLO counts	measured
Construction excess (+2 dB at 56 m)	our 32 near-site measurements	calibrated
Vehicles as sound sources	not identifiable in our data	excluded
Traffic-volume scenario	10 log10 of the volume factor	physical law

6.2 Scenario controls

Hour of day (5h-21h, each hour predicted separately, vehicle numbers follow the traffic measured at that hour) · traffic volume (x0.2 to x3) · construction on/off with working hours · mitigation: 30 km/h zone (-3 dB) or pedestrianisation (traffic x0.2).

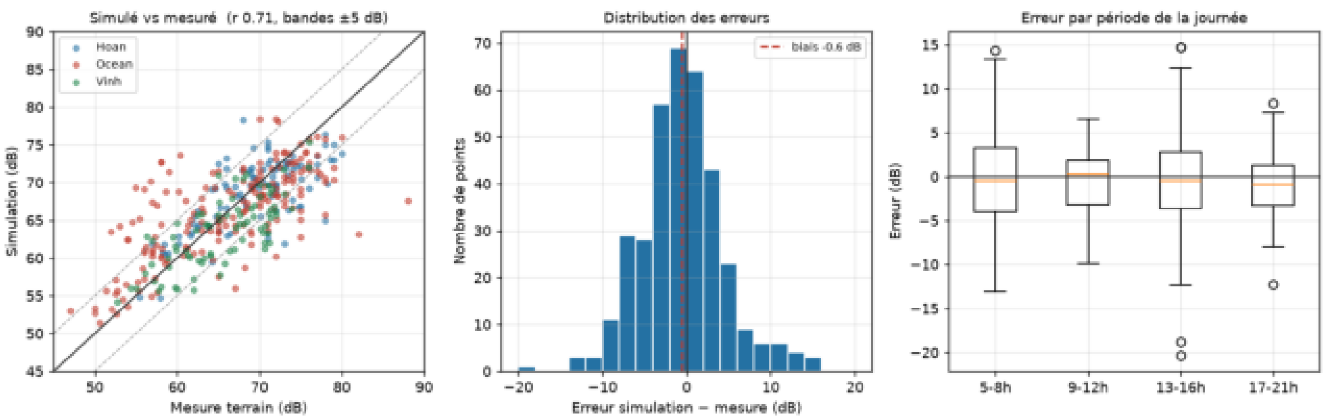
Calibration note: we tried to derive a per-vehicle emission level from our own data (non-negative energy regression on the 147 matched videos). All three coefficients came out at zero - at a given site, the number of visible vehicles does not explain the measured level (R2 0.008-0.042). Rather than inject invented values, vehicles are displayed as a calibrated picture of the fleet but carry no sound in the computation.

6.3 Does the simulation reproduce our measurements?

Metric	Value
Measurements compared	360
Bias (simulated - measured)	-0.58 dB
Mean absolute error	3.79 dB
Correlation r	0.71
Within +/- 5 dB	73 %

Each field measurement is compared with the grid cell it falls in (median 16 m away), at the hour it was taken.

Caveat: this is an in-sample check - the model that produces the grid was trained on these same points. It measures the fidelity of the chain model -> 40 m grid -> GAMA, not generalisation. The generalisation figures are those of section 5: R2 0.25 / MAE 5.01 dB under buffered loo 300 m.



7. Limitations, stated plainly

- THE LEARNED COMPONENT DOES NOT SURVIVE AN HONEST SPATIAL SPLIT. We built the hybrid architecture we recommend (physical core + LightGBM on the residual). It leads under the permissive 600 m block split (0.395 against 0.255 for the bare core) and LOSES under both strict ones (dR2 -0.123 under buffered loo 300 m). The delivered model is therefore the bare 3-parameter physical core; the residual is trained but NOT applied. Morphology aggregated over 300 m adds no measurable value over the two distance terms alone.

- ABSOLUTE CALIBRATION. The three phones are calibrated against each other, never against a reference instrument. A bias common to all three (+3.0 to +7.3 dB, bounded by anchoring on instrumented Vietnamese campaigns) is invisible in our data. The field campaign is closed and this cannot be repaired retrospectively. Contrasts are supported; absolute levels are indicative.

- MEASUREMENT QUANTITY. A 20-30 s sample is not the L_Aeq of any regulatory reference period. A single horn burst moves it by several dB - horn events reach +17 dB in Vietnamese traffic. This variance is a property of the quantity and caps the R^2 any spatial model can reach.

- GENERALISATION. With three sampled typologies, the number of independent morphological configurations available is close to three, whatever the number of points. The delivered physical model extrapolates markedly better than the learned ones (section 5.3), but three typologies remain three: maps are clipped to the sampled envelope regardless of the score.

- SPATIAL RESOLUTION. Features are aggregated over a 300 m radius, so adjacent 40 m cells share more than 98% of their disc. The map cannot resolve the facade/courtyard contrast (10-15 dB in dense fabric): predicted levels are visibly flatter than measured ones.

- TEMPORAL COVERAGE. Night (21:00-06:00) holds ~3% of measurements and nothing between 00:00 and 05:00, although that is the period with the strictest threshold. One season only.

- TRAFFIC VIDEOS. Counts are now real FLOW (tracking + line crossing), but speed is still unobservable without a ground homography and the field of view is not georeferenced, so distance is discarded. The detector was never validated against manual counts: modal splits remain lower bounds on motorcycle share (section 7).

8. Next steps

- RICHER PHYSICS, NOT MORE LEARNING. Our 3-parameter line-source core already beats every learned variant. The next gain should come from a real propagation model (CNOSSOS-EU via NoiseModelling: screening, reflections, canyon geometry), not from more capacity fitted to 363 points. A learned residual should be re-tested only once the sample covers more typologies - on this one it degrades generalisation.

- TRAFFIC. Flow by class is now measured; SPEED is the missing half. Estimate it from track displacement under a ground homography, which also georeferences the field of view and so restores the source-receiver distance. Validate the detector against manual counts first.

- MORE TYPOLOGIES. Three sampled districts bound every generalisation claim we can make. A fourth and fifth contrasting fabric would do more than any modelling change.

- DATA PUBLICATION. Deposit measurements, forms and code with a DOI; add an ethics statement covering the public-space video recordings.

9. Summary

363 smartphone measurements across 3 districts document a consistent exposure pattern in space and time. The delivered model is a 3-parameter PHYSICAL line-source law: R^2 0.25 [0.10, 0.34] / MAE 5.01 dB under buffered loo 300 m, ahead of a log(distance) regression (0.20), a 6-feature LightGBM (0.14) and the physics+ML hybrid we built to improve on it (0.12).
Three negative results are contributions: every learned elaboration gains under a permissive spatial split and loses under a strict one; cross-city transfer from Uganda fails; and video counts yield no emission coefficient, as density OR as flow. We claim contrasts, not absolutes.